BIO 101

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ASSIGNMENT

Questions:

1.) Define the term reproduction in unicellular and multicellular organisms.
2.) Mention the two types of reproductions in living things with examples.
3.) Write short note on the following: Cutting, grafting and budding.
4.) What is pollination? Explain the different types of pollination.

Answers:

1) **REPRODUCTIONS IN UNICELLULAR ORGANISMS** occur through binary fission. This involves a single cell dividing and giving rise to two daughter cells. WHILE **REPRODUCTION IN MULTICELLULAR ORGANISMS** occur through both asexual and sexual methods. In these organisms, two parents are needed and the offspring undergoes growth and regeneration.

2) **SEXUAL REPRODUCTION** is the process of creating new individual involving two organisms which are the male gamete and the female gamete to form their off spring, while **ASEXUAL REPRODUCTION** is the process of which a new organism is formed through binary fission involving one parent organism to form a daughter cell.

EXAMPLES OF SEXUAL AND ASEXUAL REPRODUCTIVE LIVING THINGS

ASEXUAL

- HYDRA
- BACTERIA
- BLACKWORM

SEXUAL

- HUMAN
- FLOWERING PLANTS
- INSECTS
- MAMMALS
- FISHES

3) i. **CUTTING**: a cutting is a section of plants such as modified stem leaf or root used for vegetative propagation that forms either adventitious shoot or adventitious roots

ii. **GRAFTING** is an artificial method of asexual reproduction used to produce plants combining favourable stems, Characteristics with favourable root characters. The stem of the plant is known as "**SCION**" while the root is called the "**STOCK**".

iii. **BUDDING** is a form of asexual reproduction in which a new individual is develops from generative anatomical part of the parent organism.

3) **POLLINATION** can be refers to the transfer of pollen grains from the male anther of a flower to the female stigma. The types of pollinations are self- pollination and cross- pollination.

   SELF-POLLINATION: it is referred to as the primary type of pollination which occurs by transferring the pollen grain directly from the anther of the male flower into the stigma of the same female flower

   CROSS-POLLINATION: It is referred to as the complex type of pollination during which the pollen grain are transferred from the anther of one flower to the stigma of another flower.